

## **Lab 5: State Diagram**

### ***Smart Elevator Control System***

A smart elevator control system manages elevator movement in a multi-storey building while ensuring passenger safety. Initially, the elevator remains idle until a floor request is received. Once a request is made, the elevator begins moving toward the destination floor. During movement, the system must simultaneously monitor safety conditions such as overload detection and door obstruction sensors.

If an unsafe condition is detected, the elevator halts movement until the issue is resolved.

Once the destination floor is reached and all safety conditions are satisfied, the elevator doors open automatically. The doors remain open for a fixed duration before closing and returning the elevator to the idle state.